

Project Name: SafeLink

Event: STARK Hackathon 2026

Team Name: Andalus

Team Members: Hayat Nura, Marya Tawfiq, Nuha Samson, Sifen Beyan

Tech Stack: ReactJS, NodeJS, ExpressJS, Tailwind CSS, MongoDB

Project Status: Hackathon Prototype

1. Executive Summary

Imagine experiencing sexual harassment or violence and reaching a point where you need help—but you do not know where to start. While hospitals, counselors, and support organizations exist, knowing that help exists is not the same as knowing how to safely access it. For someone who is already afraid, overwhelmed, or uncertain, navigating fragmented services alone becomes a major barrier.

SafeLink is a privacy-first, voice-enabled navigation platform designed to bridge this gap. It does not replace professionals; instead, it serves as an accessible entry point that guides survivors through their options and safely connects them to the right human and professional support services.

2. Problem Statement

People affected by sexual and gender-based violence (SGBV) frequently struggle to find reliable information, understand their options, and connect with appropriate care. The core issue is often not a total absence of support services, but rather the severe difficulty survivors face when trying to take the first step.

Key Barriers Addressed:

- **Information Fragmentation:** Difficulty identifying which specific service (medical, legal, or psychological) matches their immediate needs.
 - **Privacy & Stigma:** Fear of exposure, judgment, or retaliation when seeking information openly.
 - **Accessibility & Literacy:** Barriers caused by limited digital literacy, geographic distance, or a lack of tools optimized for sensitive situations.
 - **Language Barriers:** A lack of support platforms available in native local languages.
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3. Ideation History & Rejected Concepts

this section traces our team's decision-making process, highlighting the ideas we explored, shortlisted, and ultimately rejected to arrive at our final solution.

3.1 Health Tracking for People with Diabetes and High Blood Pressure

- **The Idea:** A mobile/web application helping chronic patients log and monitor blood glucose and blood pressure trends over time.
- **Why We Considered It:** These are critical, prevalent health conditions in Ethiopia where structured digital data logs could improve patient-doctor consultations.
- **Why We Rejected It:** This concept merely digitalizes an existing daily routine rather than solving a systemic barrier to accessing care. Achieving meaningful utility required

complex clinical integration, legal compliance for sensitive medical metrics, and long-term diagnostic tracking that could not be validated convincingly within a short hackathon timeline.

3.2 Voice-Controlled Device Interaction for Blind Users

- **The Idea:** A voice-first smartphone utility that verbalizes caller information and enables blind or visually impaired individuals to manage device controls hands-free.
- **Why We Considered It:** Standard touchscreen interfaces rely heavily on visual layouts, creating substantial accessibility friction for visually impaired individuals.
- **Why We Rejected It:** The scope was too broad to narrow down into a single, cohesive problem for a fast-paced hackathon. Building this effectively requires deep OS-level accessibility API hooks, intricate system permissions, and flawless ambient speech processing. We wanted to pivot toward a solution where tech removes an immediate, highly urgent human crisis barrier rather than building system utilities.

3.3 Voice-Based Marketplace Access for Ethiopian Farmers

- **The Idea:** A local-language voice platform connecting rural farmers directly to regional buyers to access crop pricing transparency without text dependencies.
- **Why We Considered It:** It tackled the twin challenges of low digital literacy and middleman exploitation in agricultural commerce.
- **Why We Rejected It:** Most farmers are highly likely to have digital illiteracy that's the barrier for this idea.

3.4 The Pivot to SafeLink

Analyzing our "Idea Graveyard" helped us discover our true North Star. We wanted a problem where:

1. The user faces extreme vulnerability at a single, critical moment.
2. Help exists, but the user is psychologically or physically barred from navigating it.
3. Voice interaction bypasses language, literacy, and stress-induced friction.
4. Absolute privacy is a functional necessity.

This realization crystallized the vision for SafeLink. The core bottleneck wasn't that emergency services were missing, but that **taking the first step was too terrifying and confusing**.

4. The SafeLink Solution & Core Features

SafeLink acts as a compassionate navigation and connection layer, turning a fragmented, overwhelming network of services into a clear, linear path:

Voice/Text Interaction (Voxide)→Contextual Assessment→Tailored Guidance→Secure Professional Handoff
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- **Voice-First Accessibility (Voxide Native):** Fully compliant with STARK Rule #2, our platform natively leverages voice interfaces. Typing out details of a traumatic event under distress is a massive emotional and logistical barrier. Speech breaks down this wall.
- **Ethiopian Local-Language Integration:** The voice flow natively supports **Amharic** and **Afaan Oromoo (Oromiffa)**, targeting deep, equitable localized accessibility.
- **Dynamic Resource Routing:** Guides users to verified, close-proximity physical shelters, legal aid networks, and clinical support institutions.
- **Privacy-First Guardrails & Quick Exit:** Zero non-essential user data collection protects individual anonymity. A prominent, ubiquitous "Quick Exit" button wipes the UI immediately if a user's physical privacy or safety is suddenly threatened.

Scope Boundary (What SafeLink is NOT):

SafeLink is strictly a **navigation, educational, and routing interface**. It does not provide medical diagnoses, offer legal legal counsel, investigate offenses, or replace professional crisis responders. It is designed simply to deliver users safely to human experts.

5. System User Journey

1. **Secure Landing:** The user opens the SafeLink web app (planned for hosting deployment via **EthioDeploy** for optimal performance).
 2. **Interface Modality Selection:** The user selects their language (Amharic/Afaan Oromoo/English) and chooses between voice or text input.
 3. **Voice/Text Assessment:** The user speaks or writes their immediate concern. SafeLink processes the input contextual needs without capturing identifying metrics.
 4. **Actionable Mapping:** The system populates direct contact pathways and operating parameters for relevant local services (e.g., medical, counseling, or emergency legal aid).
 5. **Secure Handoff & Session Purge:** The user establishes contact with a human advisor or saves their route. They can trigger the "Quick Exit" anytime to completely erase active session memory traces.
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6. Validation Targets & Success Metrics

Throughout the hackathon timeline, Team Andalus is actively validating these key architectural hypotheses:

- Interactive conversational routing yields lower dropout rates than static informational lists.
- Voice-driven UX dramatically scales up usage accessibility for individuals under high cognitive stress or with low digital literacy.
- Strict privacy safeguards and visible exit hooks directly correlate with increased user comfort and extended engagement loops.

Our Practical Target: Moving a user from an initial state of *"I am compromised, isolated, and completely lost"* to a state of *"I am safe, anonymous, and I know exactly where to take my next step toward healing."*